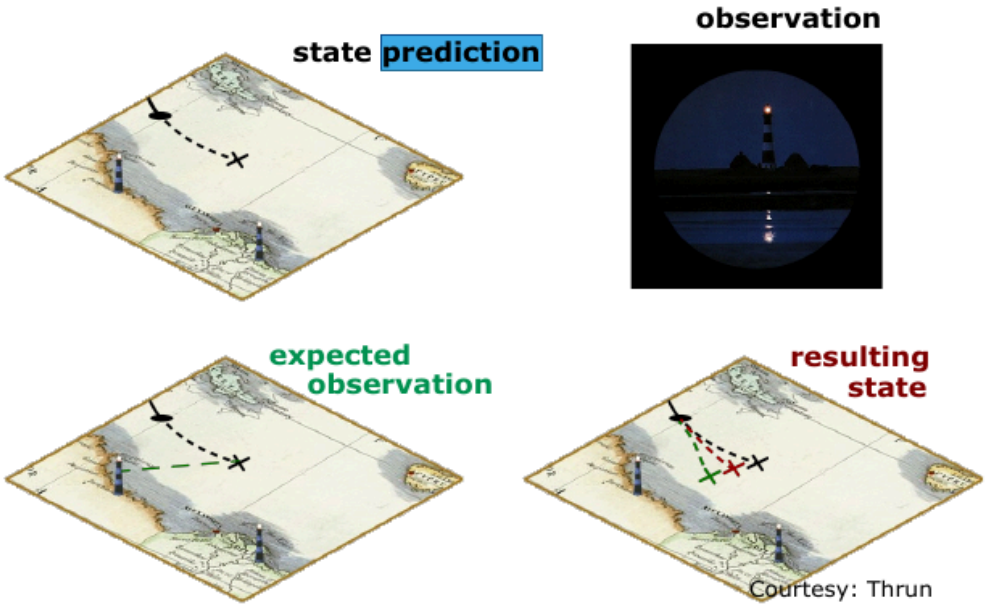
Ro08: Kalman Filter and Extended Kalman Filter (EKF)

- **Bayes filter** for the **Gaussian linear case**
  - Every distribution is Gaussian
  - $p(x) = det(2\pi\Sigma)^{-\frac{1}{2}} exp(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu))$
- Optimal solution for linear models and Gaussian distributions
- Performs recursive state estimation
- Prediction step to exploit the controls
- Correction step to exploit the observations

$$bel(x_t) = \eta \cdot \underbrace{p(z_t|x_t)}_{correction(observation)} \int \underbrace{p(x_t|x_{t-1}, u_t)}_{prediction(control)} \cdot \underbrace{bel(x_{t-1})}_{\text{recursive belief}} \, dx_{t-1}$$



Math: Updating a Gaussian Belief

- Given:
  - $x = \begin{bmatrix} x_a \\ x_b \end{bmatrix}$
  - $p(x) = \mathcal{N}$
- Marginals are Gaussians
  - $p(x_a) = \mathcal{N}$
  - $p(x_b) = \mathcal{N}$
- Conditionals are Gaussians
  - $p(x_a|x_b) = \mathcal{N}$
  - $p(x_b|x_a) = \mathcal{N}$
- Given $p(x) = p(x_a, x_b) = \mathcal{N}(\mu, \Sigma)$ with:
  - $\mu = \begin{bmatrix} \mu_a \\ \mu_b \end{bmatrix}$
  - $\Sigma = \begin{bmatrix} \Sigma_{aa} & \Sigma_{ab} \\ \Sigma_{ba} & \Sigma_{bb} \end{bmatrix}$

- The **marginalisation distribution** is
  - $p(x_a) = \int p(x_a, x_b) dx_b = \mathcal{N}(\mu, \Sigma)$
    - with $\mu = \mu_a$ and $\Sigma = \Sigma_{aa}$
- The **conditional distribution** is:
  - $p(x_a|x_b) = \frac{p(x_a, x_b)}{p(x_b)} \mathcal{N}(\mu, \Sigma)$
    - with $\mu = \mu_a + \Sigma_{ab}\Sigma_{bb}^{-1}(x_b - \mu_b)$
    - $\Sigma = \Sigma_{aa} - \Sigma_{ab}\Sigma bb^{-1}\Sigma_{ba}$

Linear Models for Motions and Observations

- Both models expressable through a linear function
  - $f(x) = Ax + b$
- A Gaussian that is transformed through a linear function stays Gaussian
- Kalman filter assumes a linear transition and observation model
- Noise is a zero-man Gaussian

$$x_t = A_t x_{t-1} + B_t u_t + \epsilon_t$$
$$z_t = C_t x_t + \delta_t$$

Components of Kalman Filter

- A_t: Matrix $(n \times n)$ that describes how the state evolves from $t-1$ to t without controls or noise
- B_t: Matrix $(n \times l)$ that describes how the control u_t changes the state form $t-1$ to t
- C_t: Matrix $(k \times n)$ that describe mapping of state x_t to an observer z_t
- ϵ_t & δ_t: Random variables representing process and measurement noise
  - assumed to be independent and normally distributed with covariance R_t and Q_t respectively

Linear Motion Model

- Motion under Gaussian noise leads to

$$p(x_t|u_t, x_{t-1}) = det(2\pi R_t)^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(x_t - A_t x_{t-1} - B_t u_t)^T R_t^{-1}(x_t - A_t x_{t-1} - B_t u_t)\right)$$

- R_t describes the noise of the motion
- Note: swap in literature - $R_t \leftrightarrow Q_t$

Linear Observation Model

- Measuring under Gaussian noise leads to

$$p(z_t|x_t) = det(2\pi Q_t)^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(z_t - C_t x_t)^T Q_t^{-1}(z_t - C_t x_t)\right)$$

- Q_t describes the measurement noise

Everything Stays Gaussian

- Given initial Gaussian belief, the belief stays Gaussian
- $\overline{bel}(x_t) = \int \underbrace{p(x_t|u_t, x_{t-1})}_{Gaussian} \quad \underbrace{bel(x_{t-1})}_{Gaussian} dx_{t-1}$
- Convolution of two Gaussians is a Gaussian

- We can write

$$\begin{aligned}\overline{bel}(x_t) \\ = \quad \eta \int \exp\left(-\frac{1}{2}\,(x_t - A_t\,x_{t-1} - B_t\,u_t)^\top\,R_t^{-1}\,(x_t - A_t\,x_{t-1} - B_t\,u_t)\right) \\ \exp\left(-\frac{1}{2}\,(x_{t-1}-\mu_{t-1})^\top \Sigma_{t-1}^{-1}(x_{t-1}-\mu_{t-1})\right)\,dx_{t-1}\end{aligned}$$

- and thus

$$\begin{aligned}\overline{bel}(x_t) &= \eta \int \exp\left(-L_t\right)\,dx_{t-1} \\ L_t &= \frac{1}{2}\,(x_t - A_t\,x_{t-1} - B_t\,u_t)^\top\,R_t^{-1}\,(x_t - A_t\,x_{t-1} - B_t\,u_t) \\ &\quad + \frac{1}{2}\,(x_{t-1}-\mu_{t-1})^\top\,\Sigma_{t-1}^{-1}\,(x_{t-1}-\mu_{t-1})\end{aligned}$$

- We can split up L_t so that one part that depends on x_t and on x_t, x_{t-1}

$$L_t = L_t(x_{t-1}, x_t) + L_t(x_t)$$

- Thus

$$\begin{aligned}\overline{bel}(x_t) &= \eta \int \exp\left(-L_t(x_{t-1}, x_t) - L_t(x_t)\right)\,dx_{t-1} \\ &= \eta\,\underbrace{\exp\left(-L_t(x_t)\right)}_{\text{Gaussian}}\,\underbrace{\int \exp\left(-L_t(x_{t-1}, x_t)\right)\,dx_{t-1}}_{\text{Marginalization}}\end{aligned}$$

- Given initial Gaussian belief, the belief stays Gaussian

$$\underbrace{\overline{bel}(x_t)}_{\text{Gaussian}} = \underbrace{\int \underbrace{p(x_t \mid u_t, x_{t-1})}_{\text{Gaussian}}\,\underbrace{bel(x_{t-1})}_{\text{Gaussian}}\,dx_{t-1}}_{\text{Gaussian}}$$

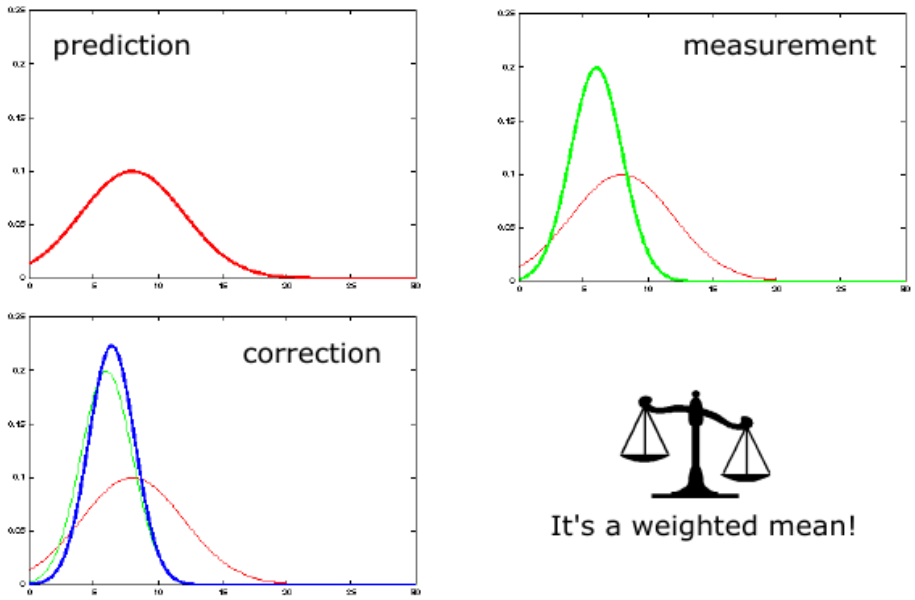
$$\underbrace{bel(x_t)}_{\text{Gaussian}} = \eta\,\underbrace{p(z_t \mid x_t)}_{\text{Gaussian}}\,\underbrace{\overline{bel}(x_t)}_{\text{Gaussian}}$$

Everything is and stays Gaussian!

To Derive Kalman Filter Algorithm One Exploits

- Product of 2 Gaussians is a Gaussian
- Gaussians stay Gaussians
  - under linear transformations
  - Marginal and conditional distribution of a Gaussiann stay
- Computing mean and covariance of marginal and conditional of a Gaussian
- Matrix inversion lemma
- ...

$$
\begin{aligned}
&\textbf{Kalman_filter}(\mu_{t-1}, \Sigma_{t-1}, u_t, z_t)\textbf{:} \\
&\quad \bar{\mu}_t = A_t\ \mu_{t-1} + B_t\ u_t \\
&\quad \bar{\Sigma}_t = A_t\ \Sigma_{t-1}\ A_t^\top + R_t \\
&\quad K_t = \bar{\Sigma}_t\ C_t^\top (C_t\ \bar{\Sigma}_t\ C_t^\top + Q_t)^{-1} \\
&\quad \mu_t = \bar{\mu}_t + K_t(z_t - C_t\ \bar{\mu}_t) \\
&\quad \Sigma_t = (I - K_t\ C_t)\ \bar{\Sigma}_t \\
&\quad return\ \mu_t, \Sigma_t
\end{aligned}
$$



Useless/Perfect Sensor (Q_t)

- Infinite noise: $Q_t = diag(\infty)$
  - Leads to:

$$
\begin{aligned}
4: \quad & K_t = \bar{\Sigma}_t\ C_t^\top (\cancel{C_t\ \bar{\Sigma}_t\ C_t^\top + Q_t})^{-1} \quad \to 0 \\
5: \quad & \mu_t = \bar{\mu}_t + K_t(z_t - C_t\ \bar{\mu}_t)
\end{aligned}
$$

$$
5: \quad \mu_t = \bar{\mu}_t + \underbrace{K_t(z_t - C_t\ \bar{\mu}_t)}_{\to 0}
$$

$$
5: \quad \mu_t = \bar{\mu}_t
$$

    - **Oservation** will be **ignored**
- No noise: $Q_t = diag(0)$
  - Leads to

$$4: \quad K_t = \bar{\Sigma}_t\, C_t^\top (C_t\, \bar{\Sigma}_t\, C_t^\top + \cancel{Q_t})^{-1} \quad \rightarrow C_t^{-1}$$

$$5: \quad \mu_t = \bar{\mu}_t + K_t(z_t - C_t\, \bar{\mu}_t)$$

$$5: \quad \mu_t = \bar{\mu}_t + K_t(z_t - C_t\, \bar{\mu}_t)$$

$$\rightarrow C_t^{-1} z_t - \bar{\mu}_t$$

- $5: \quad \mu_t = C_t^{-1} z_t$
- **Prediction** will be **ignored**

KF Assumptions

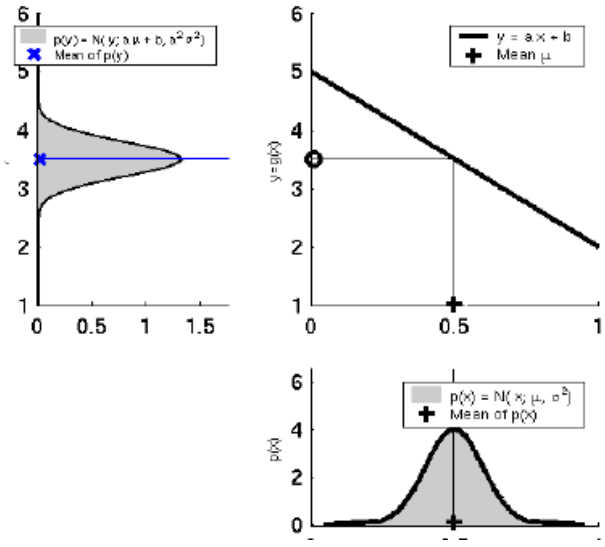
- Gaussian distributions and noise
- Linear motion and observation model
- $x_t = A_t x_{t-1} + B_t u_t + \epsilon_t$
- $z_t = C_t x_t + \delta_t$

Extended Kalman Filter

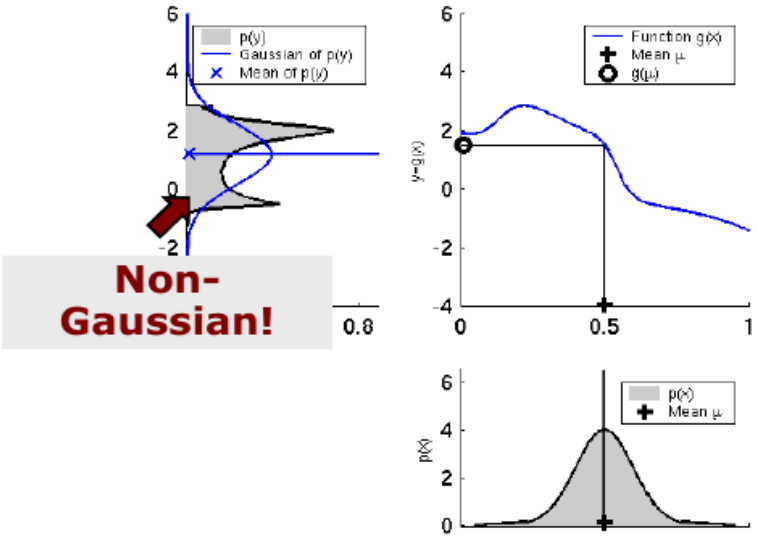
- Most realistic problems involve nonlinear functions

$$\cancel{x_t = A_t x_{t-1} + B_t u_t} + \epsilon_t \qquad \cancel{z_t = C_t x_t + \delta_t}$$

$$\Downarrow \qquad\qquad \Downarrow$$

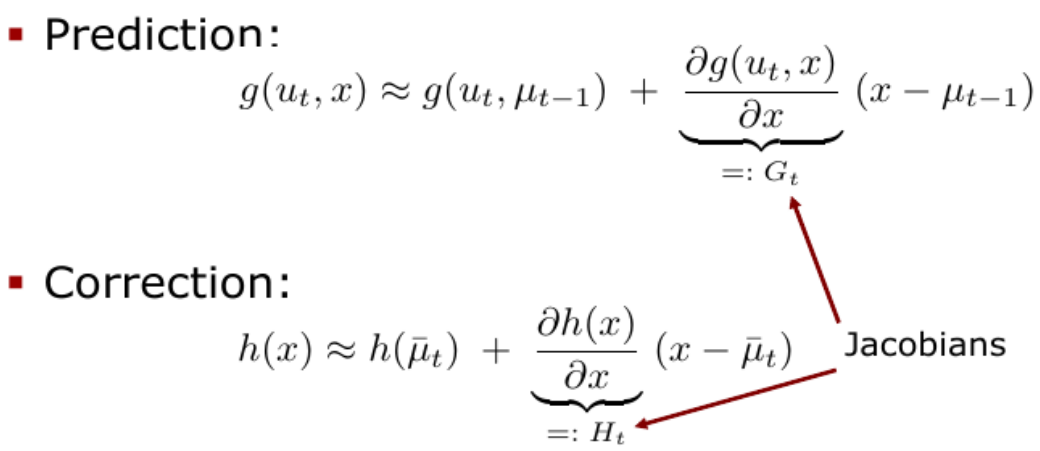
- $$x_t = g(u_t, x_{t-1}) + \epsilon_t \qquad z_t = h(x_t) + \delta_t$$
- Linearity Assumption
  - 

- Non-Linear Function

- 

  - non-linear functions lead to non-Gaussians distribution
    - -> Kalman filter is not applicable anymore
- -> Local linearisation

EKF Linearisation

- 

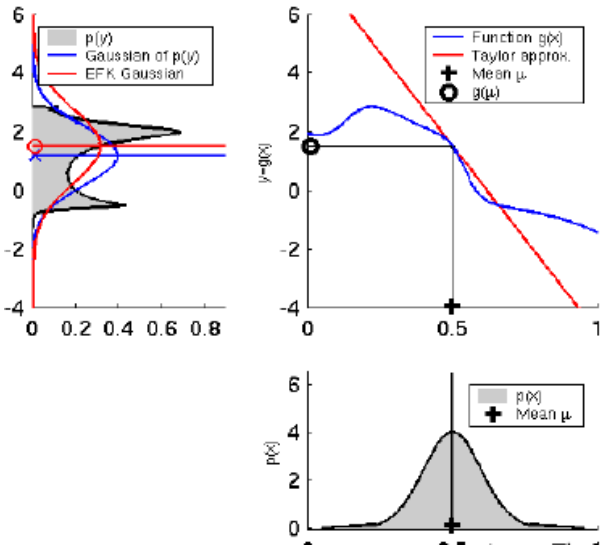
- Jacobian matrix
  - It is a non-square matrix $m \times n$ in general
  - Given a vector-valued function

$$g(x) = \begin{pmatrix} g_1(x) \\ g_2(x) \\ \vdots \\ g_m(x) \end{pmatrix}$$

  - The Jacobian is defined as

$$G_x = \begin{pmatrix} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} & \cdots & \frac{\partial g_1}{\partial x_n} \\ \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} & \cdots & \frac{\partial g_2}{\partial x_n} \\ \vdots & \vdots & \cdots & \vdots \\ \frac{\partial g_m}{\partial x_1} & \frac{\partial g_m}{\partial x_2} & \cdots & \frac{\partial g_m}{\partial x_n} \end{pmatrix}$$

  - Orientation of tangent plane to the vector-valued function at a given point
  - Generalises the gradient of a scalar valued function
  - Linear Function

- 

Linearised Motion Model

- Linearised model leads to

$$p(x_t|u_t x_{t-1}) \approx det(2\pi R_t)^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(x_t - g(u_t, \mu_{t-1}) - G_t(x_{t-1} - \mu_{t-1}))^T R_t^{-1}(x_t - \underbrace{g(u_t, \mu_{t-1}) - G_t(x_{t-1} - \mu_{t-1})}_{\text{linearised model}})\right)$$

- R_t describes the motion noise

Linearised Observation Model

- Linearised model leads to

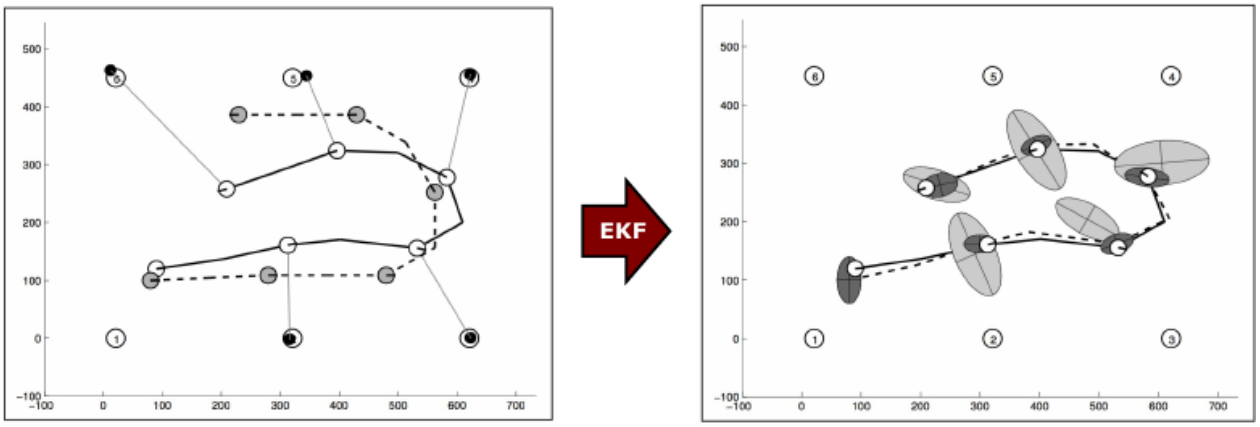
$$p(z_t|x_t) = det(2\pi Q_t)^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(z_t - h(\overline{\mu}_t) - H_t(x_t - \overline{\mu}_t))^T Q_t^{-1}(z_t - \underbrace{h(\overline{\mu}_t) - H_t(x_t - \mu_t)}_{\text{linearised model}})\right)$$

- Q_t describes the measurement noise

$$\begin{aligned}
&\textbf{Extended_Kalman_filter}(\mu_{t-1}, \Sigma_{t-1}, u_t, z_t)\textbf{:}\\
&\quad \bar{\mu}_t = g(u_t, \mu_{t-1})\\
&\quad \bar{\Sigma}_t = G_t\, \Sigma_{t-1}\, G_t^\top + R_t\\
&\quad K_t = \bar{\Sigma}_t\, H_t^\top (H_t\, \bar{\Sigma}_t\, H_t^\top + Q_t)^{-1}\\
&\quad \mu_t = \bar{\mu}_t + K_t(z_t - h(\bar{\mu}_t))\\
&\quad \Sigma_t = (I - K_t\, H_t)\, \bar{\Sigma}_t\\
&\quad \textit{return } \mu_t, \Sigma_t
\end{aligned}$$

prediction

correction



Weighted sum of predictions and observations

KF Summary

- Kalman filter is realisation of Bayes filter for the linear Gaussian case
- All models/beliefs must be Gaussian
- Linear motion and observation model
- Optimal estimator under these assumptions
- Linear models are a serious limitation for real-world applications

EKF Summary

- Way to handle non-linearities
  - local linearisation via Taylor expansion
- Works well in practice for **moderate non-linearities**
- Large uncertainties in the beliefs leads to increased approximation errors